Pythagoras

Geometry · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 108 minutes · Total marks: 108

KEY RULES

Theorem	$a^2 + b^2 = c^2$ (c is the hypotenuse, opposite the right angle)
Hypotenuse	Add the squares, then square root
Short side	Subtract the squares, then square root
3D	Use two right-angled triangles in turn
Distance	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

WORKED EXAMPLE 1

[2 marks]

A right-angled triangle has shorter sides 6 cm and 8 cm. Work out the hypotenuse.

Step 1	$c^2 = a^2 + b^2$	$c^2 = 36 + 64 = 100$
Step 2	Square root	$c = 10$

Answer: 10 **cm**

WORKED EXAMPLE 2

[3 marks]

A right-angled triangle has hypotenuse 13 cm and one side 5 cm. Work out the other side.

Step 1	$a^2 = c^2 - b^2$	$a^2 = 169 - 25 = 144$
Step 2	Square root	$a = 12$

Answer: 12 **cm**

WORKED EXAMPLE 3

[3 marks]

Work out the distance between $A(-2, 1)$ and $B(4, 5)$. Give your answer to 3 significant figures.

Step 1	Horizontal and vertical differences	6 and 4
Step 2	Pythagoras	$d^2 = 36 + 16 = 52$
Step 3	Square root	$d = 7.211\ldots$

Answer: 7.21

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Pythagoras

1 Is a triangle with sides 7 cm, 24 cm and 25 cm right-angled?

Show your working.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

2 Is a triangle with sides 5 cm, 8 cm and 10 cm right-angled?

Show your working.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

3 A rectangle is 10 cm by 24 cm. Work out the length of its diagonal.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

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4 Here is a right-angled triangle.

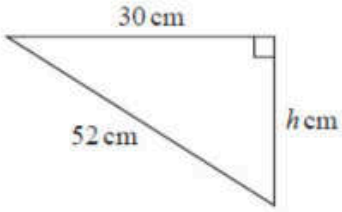


Diagram **NOT** accurately drawn

Calculate the value of h.
Give your answer correct to 3 significant figures.

h = ...

(Total for question = 3 marks)

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5 Here is a right-angled triangle.

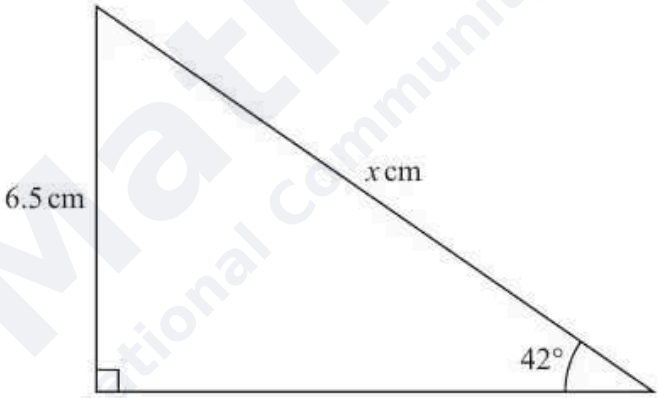


Diagram **NOT** accurately drawn

Work out the value of x.
Give your answer correct to one decimal place.

x = ...

(Total for Question 7 is 3 marks)

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6 The diagram shows a right-angled triangle.

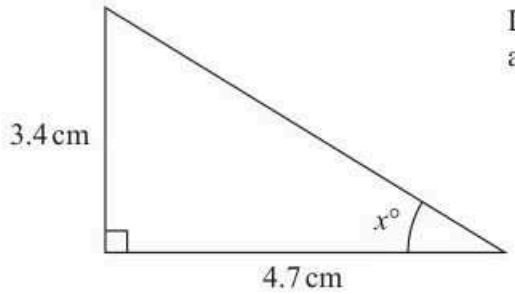


Diagram **NOT** accurately drawn

Calculate the value of x.
Give your answer correct to one decimal place.

$x =$ .....

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7

8 Here is a right-angled triangle.

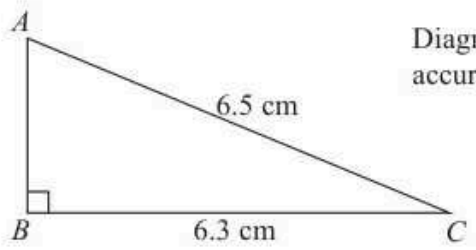


Diagram **NOT** accurately drawn

$AC = 6.5$ cm.
$BC = 6.3$ cm.
Angle $ABC = 90°$

Calculate the length of AB.

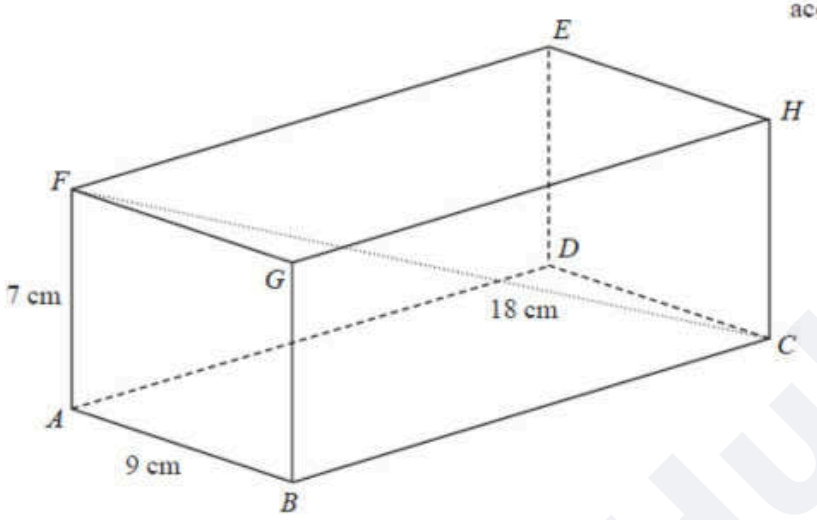
..... cm

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8 The diagram shows cuboid *ABCDEFGH*

Diagram **NOT** accurately drawn



AB = 9 cm *AF* = 7 cm *FC* = 18 cm

Calculate the length of *BC*
Give your answer correct to 3 significant figures.

..... cm

(Total for question = 3 marks)

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9 A right-angled triangle has sides x, $x+1$ and $x+2$ (the hypotenuse is $x+2$).

Show that $x^2 - 2x - 3 = 0$ and work out x.

(Total for question = 4 marks)

Practice question written for this worksheet (not from a past paper)

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10 Markus makes a steel framework.
The framework is in the shape of the right-angled triangle ABC shown in the diagram.

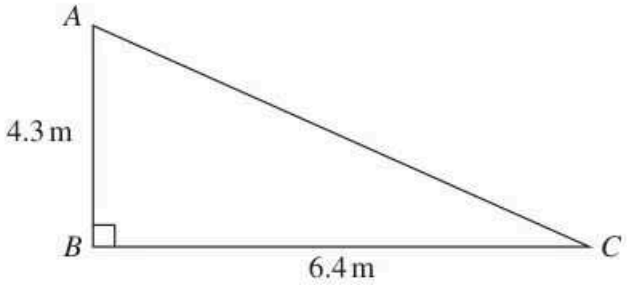


Diagram **NOT** accurately drawn

The steel that Markus uses costs \$22 per metre.
The steel can **only** be bought in a length that is a whole number of metres.

Work out the total cost of the steel that Markus buys in order to make the framework.

\$ .....

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11 The diagram shows a shaded shape *AEBCD* made by removing triangle *AEB* from rectangle *ABCD*

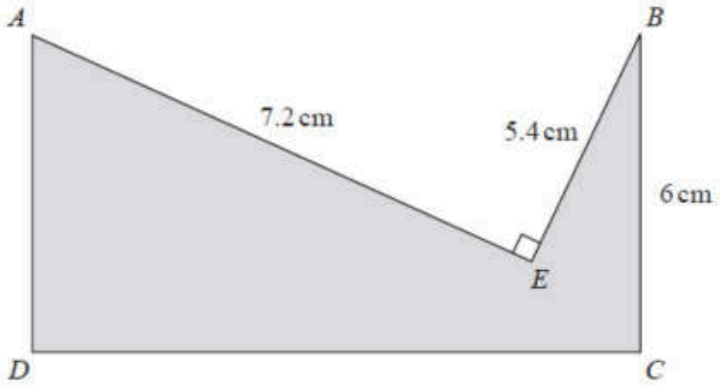


Diagram **NOT** accurately drawn

AE = 7.2 cm *BE* = 5.4 cm *BC* = 6 cm angle *AEB* = 90°

Work out the perimeter of the shaded shape.

..... cm

(Total for question = 4 marks)

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12 **11** The diagram shows a right-angled triangle ABC

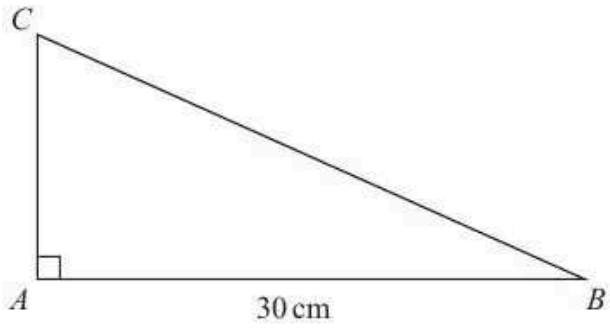


Diagram **NOT** accurately drawn

Area of triangle $ABC = 240\,\text{cm}^2$

Work out the perimeter of triangle ABC

..... cm

(Total for Question 11 is 4 marks)

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13 The diagram shows a roof support.

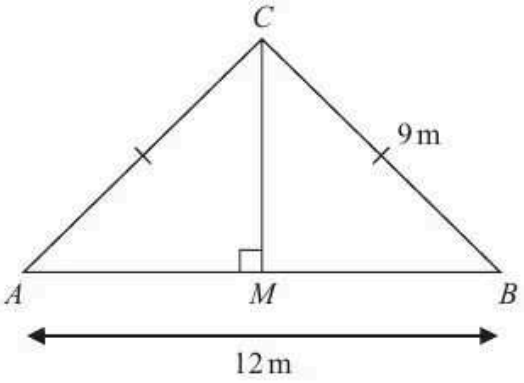


Diagram **NOT** accurately drawn

The roof support is made from four lengths of wood, AB, AC, BC and MC

$AC = BC = 9\,\text{m}$ $AB = 12\,\text{m}$

angle $AMC = 90°$

Lewis is going to buy lengths of wood to make the roof support.

The wood costs 21.50 euros per metre.
Each length of wood he buys has to be a whole number of metres.

Work out the total cost of the wood Lewis needs to buy.
Show your working clearly.

..... euros

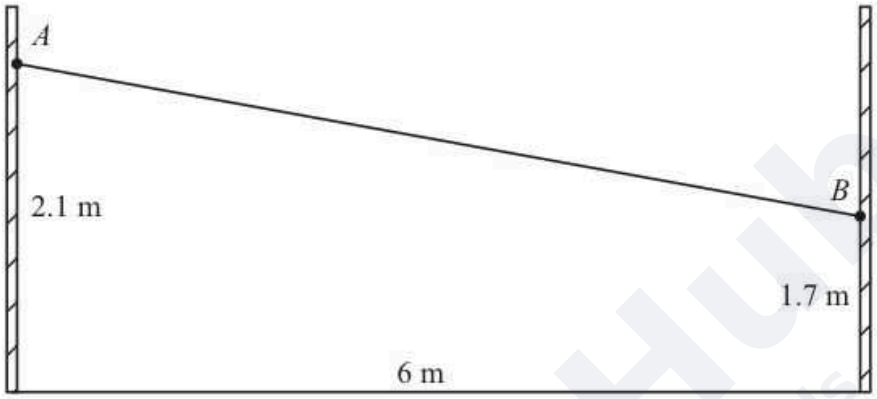
(Total for Question 11 is 4 marks)

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14 A washing line is attached at points A and B on two vertical posts standing on horizontal ground.
Point A is 2.1 metres above the ground on one post.
Point B is 1.7 metres above the ground on the other post.
The horizontal distance between the two posts is 6 metres.

Diagram **NOT** accurately drawn



Calculate the distance AB.
Give your answer correct to 3 significant figures.

..... m

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Pythagoras

15 The diagram shows an isosceles triangle ABC

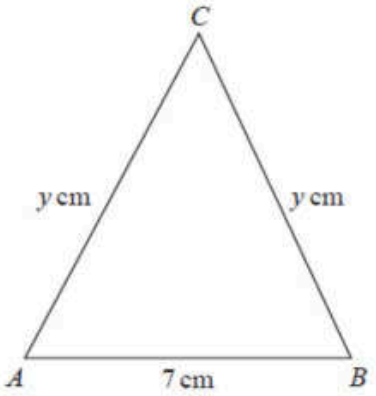


Diagram **NOT** accurately drawn

$AB = 7$ cm $AC = BC = y$ cm

The area of the triangle is 42 cm^2
Work out the value of y

$y =$ .....

(Total for question = 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · January 2023 · Question 24

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16 Two circles, C_1 and C_2 , are drawn on a centimetre grid, with a scale of 1 cm for 1 unit on each axis.

The centre of circle C_1 is at the point with coordinates (–1, 3) and the radius of C_1 is 13 cm.

The centre of circle C_2 is at the point with coordinates (7, 18) and the radius of C_2 is 6 cm.

(a) Work out the distance between the centre of C_1 and the centre of C_2

..... cm

(3)

(b) Explain why circle C_1 intersects circle C_2

.....

.....

(1)

(Total for question = 4 marks)

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17 The diagram shows a cube $ABCDEFGH$ with sides of length 6 cm.

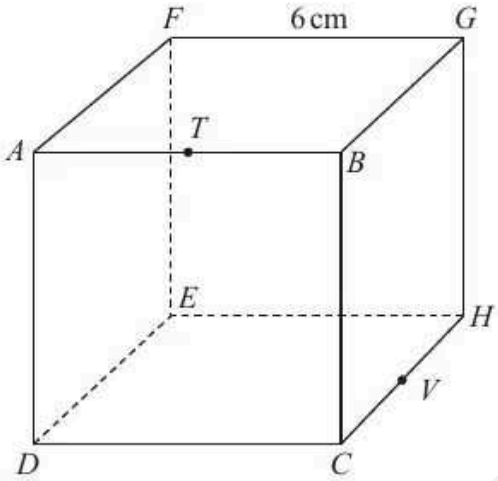


Diagram **NOT** accurately drawn

T is the midpoint of AB and V is the midpoint of CH

Work out the distance from T to V in a straight line through the cube.
Give your answer in the form $\sqrt{a}$ cm where a is an integer.

..... cm

(Total for Question 18 is 4 marks)

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18 **23**

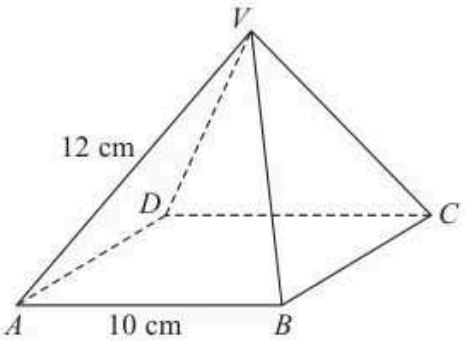


Diagram **NOT** accurately drawn

ABCD is the square base of the pyramid *VABCD*.

$AB = BC = CD = DA = 10$ cm.
$VA = VB = VC = VD = 12$ cm.

Calculate the height of the pyramid.
Give your answer correct to 3 significant figures.

..... cm

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19 The diagram shows a right-angled triangle.

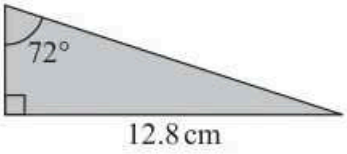


Diagram **NOT** accurately drawn

Five of these triangles are put together to make a shape.

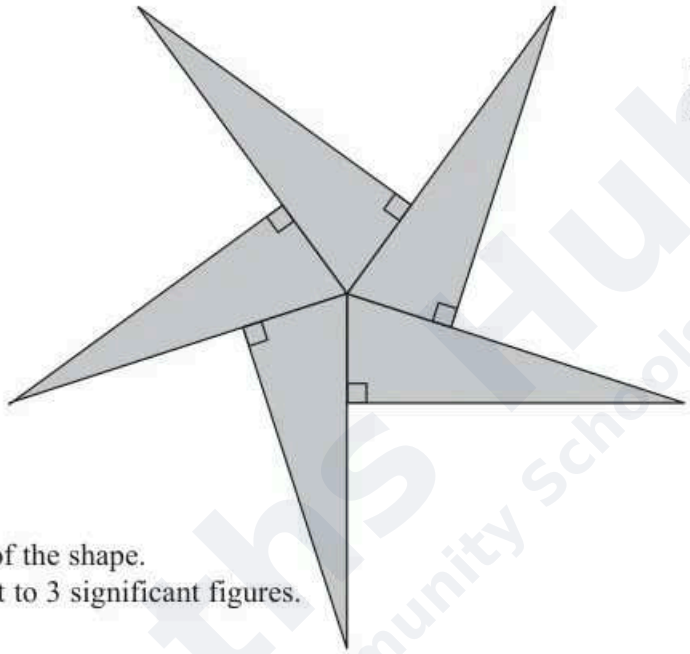


Diagram **NOT** accurately drawn

Calculate the perimeter of the shape.
Give your answer correct to 3 significant figures.

..... cm

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20 **10** ABD and ABC are right-angled triangles.

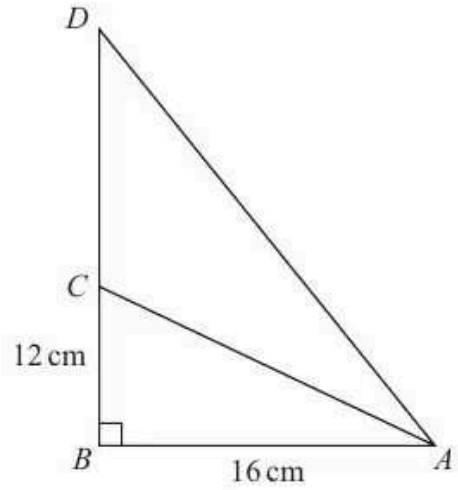


Diagram **NOT** accurately drawn

$AB = 16\,\text{cm}$ $BC = 12\,\text{cm}$ $AD = 1.5 \times AC$

Find the length of CD
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 10 is 5 marks)

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21 The diagram shows an isosceles triangle, with base length 24 cm.

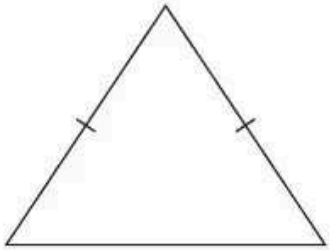


Diagram **NOT** accurately drawn

The perimeter of the triangle is 54 cm.

Work out the area of the triangle.

..... cm^2

(Total for Question 10 is 5 marks)

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22 The diagram shows a trapezium, *ABCD*

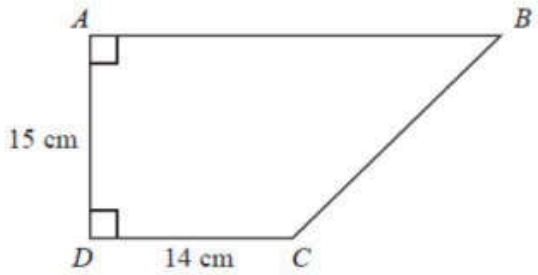


Diagram **NOT** accurately drawn

DAB and *ADC* are right angles.

AD = 15 cm *DC* = 14 cm

The area of the trapezium is 360 cm^2

Work out the perimeter of the trapezium.

..... cm

(Total for question = 6 marks)

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CHALLENGE QUESTIONS

23 The diagram shows a quadrilateral *ABCD*

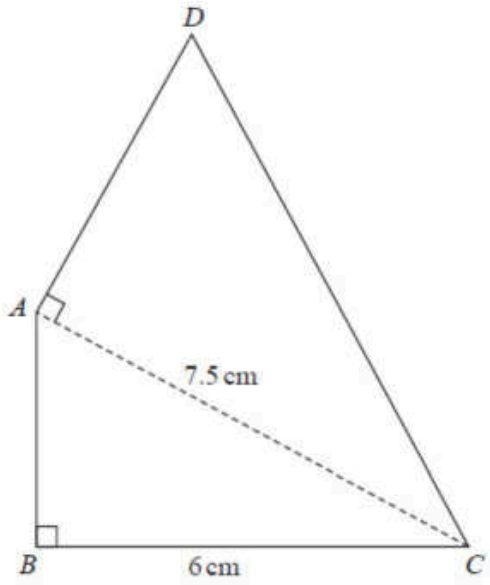


Diagram **NOT** accurately drawn

In the diagram, *ABC* and *DAC* are right-angled triangles.
BC = 6 cm *AC* = 7.5 cm

The area of quadrilateral *ABCD* is 31.5 cm^2

Work out the length of *AD*

..... cm

(Total for question = 6 marks)

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24 The diagram shows a shape made up of three semicircles, enclosing a right-angled triangle.

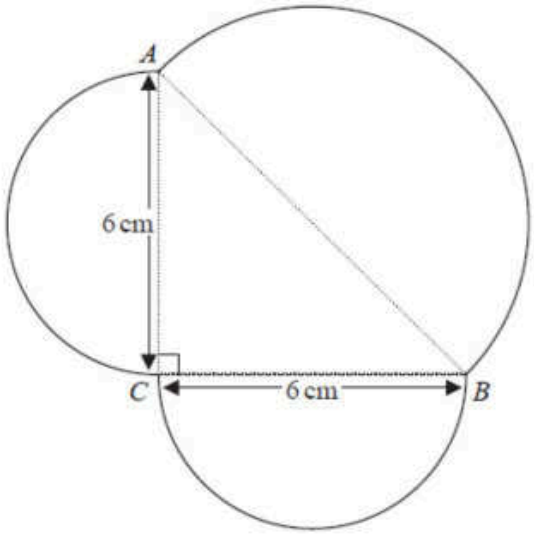


Diagram **NOT** accurately drawn

AB, *BC* and *CA* are each the diameter of a semicircle.

BC = *CA* = 6 cm.

Work out the perimeter of the shape.
Give your answer correct to one decimal place.

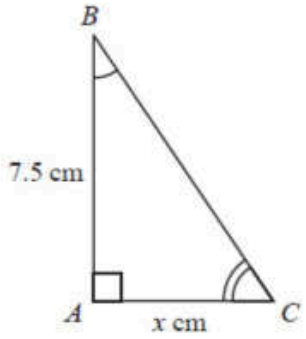
..... cm

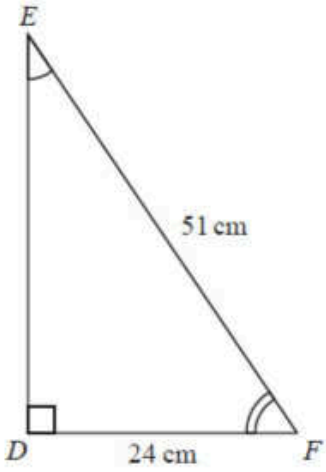
(Total for question = 5 marks)

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25 Here are two similar triangles.





Diagrams **NOT** accurately drawn

Work out the value of x
Show your working clearly.

x = ...

(Total for question = 5 marks)

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26 **11** The diagram shows two triangles, *ADE* and *CDB*

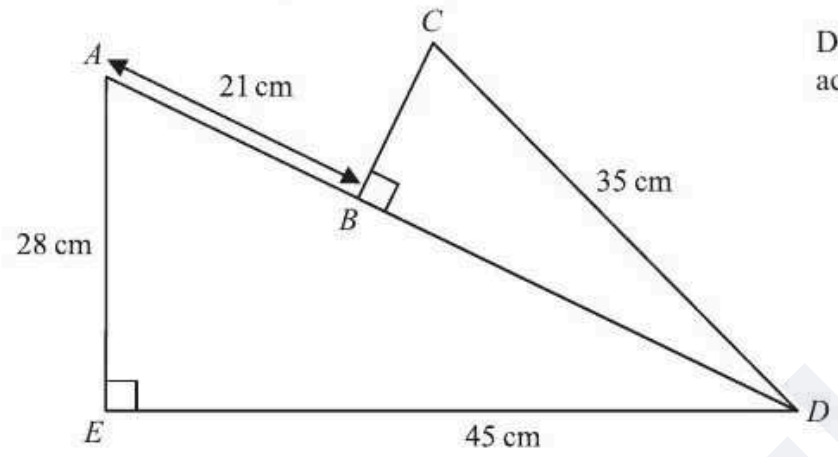


Diagram **NOT** accurately drawn

ABD is a straight line.

$AE = 28\,\text{cm}$ $ED = 45\,\text{cm}$ $AB = 21\,\text{cm}$ $CD = 35\,\text{cm}$

angle *AED* = angle *CBD* = 90°

Work out the area of triangle *CDB*
Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 11 is 5 marks)

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Pythagoras

27 The diagram shows a shaded shape *ABCDEO* made from a semicircle *BCD* and two identical right-angled triangles *ABO* and *EDO*

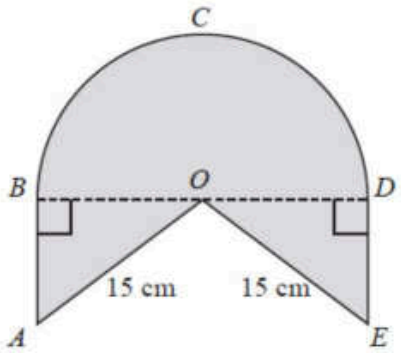


Diagram **NOT** accurately drawn

BD is the diameter of the semicircle.
O is the midpoint of *BD*

angle *ABO* = angle *EDO* = 90° *AO* = *EO* = 15 cm

The area of semicircle *BCD* is 227 cm^2

Work out the perimeter of the shaded shape.
Show your working clearly.
Give your answer correct to one decimal place.

..... cm

(Total for question = 5 marks)

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TOTAL FOR PAPER IS 108 MARKS

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